BRAIN — DentaFlow Dental Clinic Platform

Purpose

This file is the shared project brain for DentaFlow, a dental clinic management platform. It captures the product context, decisions, terminology, constraints, and current priorities so that anyone joining the project can quickly understand what we are building and why.

Product in one sentence

DentaFlow helps dental clinics manage patients, appointments, clinical records, billing, and follow-ups in one secure, easy-to-use system.

North-star outcome

Increase clinic efficiency and patient retention by reducing missed appointments, improving record completeness, and making treatment-to-payment workflows seamless.

Core users

- **Patient:** Wants convenient booking, reminders, clear costs, and post-treatment instructions.
- **Front desk:** Wants fast scheduling, check-in, payment collection, and communication tools.
- **Dentist:** Wants reliable patient history, clinical charting, treatment planning, and documentation.
- **Hygienist:** Wants simple assessment and cleaning documentation.
- **Clinic owner:** Wants visibility into revenue, utilization, no-shows, and growth.

Product principles

1. **Clinic-first workflow:** Mirror how a dental clinic operates: reception → intake → chair → treatment → billing → recall.
2. **Reduce manual work:** Automate reminders, recalls, invoices, and routine documentation.
3. **Clinical clarity:** Make patient history, alerts, and treatment plans easy to see before care begins.
4. **Trust and privacy:** Patient data must be protected, access-controlled, and auditable.
5. **Simple over exhaustive:** The MVP should remove paper and spreadsheets, not replace every specialized dental system.

Vocabulary

Term	Meaning
Patient chart	Complete patient profile including demographics, history, documents, notes, and treatments
Dental chart	Tooth-by-tooth clinical record of conditions and procedures
Treatment plan	Proposed procedures, costs, sequencing, and patient approval
Chair	Physical dental operatory/room where treatment occurs
Recall	Scheduled reminder for preventive care or follow-up
No-show	Patient misses an appointment without cancelling
Outstanding balance	Amount invoiced but not yet fully paid
Treatment conversion	Percentage of proposed treatment plans accepted by patients

Current product decisions

Decision	Rationale
Build web app first	Clinic staff primarily work from desktops; mobile apps can come later
Include online booking in MVP	Directly targets no-shows and front-desk workload
Use role-based access	Dentists, hygienists, front desk, billing, and admins need different permissions
Keep insurance claim automation out of MVP	Claim workflows vary heavily by region and payer
Start with SMS/email reminders	Highest-impact, lowest-complexity communication feature
Maintain audit trail from day one	Clinical and billing records require accountability

Known constraints

- Must support clinic-specific working hours, holidays, and provider schedules.
- Patient medical/dental history must remain current and reviewable by the dentist.
- The platform must avoid becoming a diagnostic decision-maker in MVP; it supports documentation and workflow, not automated diagnosis.
- Local healthcare privacy, consent, retention, and billing requirements must be confirmed before launch.

Open questions

- Which payment providers and currencies are required?
- Do we need multi-language patient forms at launch?
- Should the dental chart integrate with an existing imaging system?
- What is the minimum patient-data import format from the clinic's current system?
- Should patients see full invoices in the portal or only payment status?

Immediate priorities

1. Finalize appointment scheduling rules and cancellation policy.
2. Define patient intake form fields and consent flow.
3. Design dental chart and treatment-plan data model.
4. Select SMS/email and payment providers.
5. Define roles, permissions, and audit-log requirements.
6. Prepare patient-data migration template.

Definition of done

A feature is done when:

- It works for the intended user role.
- It handles validation, permissions, and error states.
- Patient-facing text is clear and accessible.
- Relevant actions are logged for audit.
- It has basic test coverage.
- Documentation and training notes are updated.

Do not build yet

- AI diagnosis or treatment recommendations
- Full insurance clearinghouse integration
- Multi-location franchise administration
- Native iOS/Android apps
- Complex inventory forecasting